Hyperbolic Functions

Overview

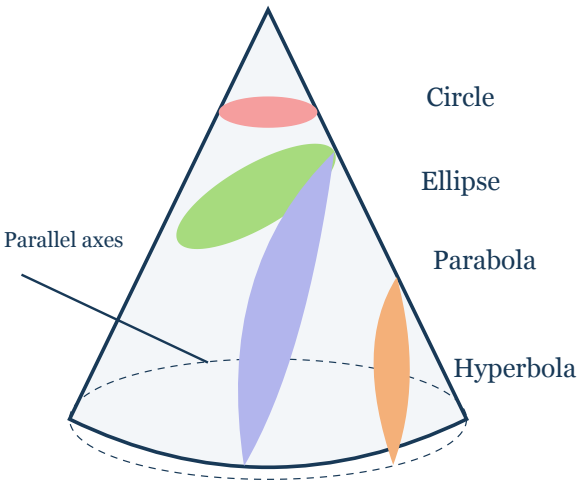
Solution 2

We will study the definition and purpose of hyperbolic functions, such as $\sinh x$ and $\cosh x$, their inverses, and how to manipulate, differentiate and integrate them.

1. Definitions and sketches: find the exact value of $\tanh(\ln 4)$.
2. Inverse functions: prove $\operatorname{arcosh} x = \ln(x + \sqrt{x^2 - 1})$.
3. Identities and equations: solve $2\cosh^2 x - 5\sinh x = 5$ for real x.
4. Differentiation: determine $\frac{d}{dx}\operatorname{arsinh}(\sqrt{x})$.
5. Integration: determine $\int \frac{1}{\sqrt{12x+2x^2}}\,dx$.

Conic Sections

Solution 3



The axis of the parabola is parallel to the side of the cone.

There are different families of curves. FP1 covers ellipses, parabolas and hyperbolas in Chapters 2 and 3, “Conic Sections I and II”.

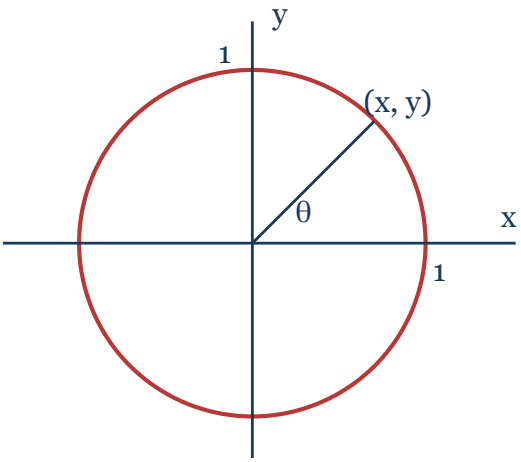
Each family has different properties and equations. Slicing a cone in different ways produces these curves, hence “conic sections”.

Comparing Circles and Hyperbolas

Solution 4

Context only — no notes needed. Hyperbolas in FP1 explain the name “hyperbolic”.

The simplest circle is the unit circle centred at the origin.



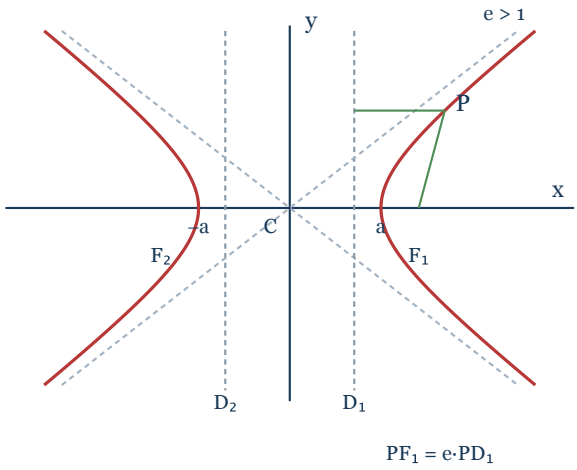
Cartesian equation:

$$x^2 + y^2 = 1.$$

Parametric equations:

$$x = \cos\theta, \quad y = \sin\theta.$$

The equivalent unit hyperbola crosses the x-axis at $(1, 0)$ and $(-1, 0)$.



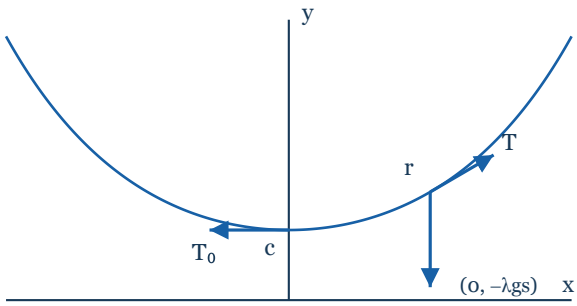
Cartesian equation:

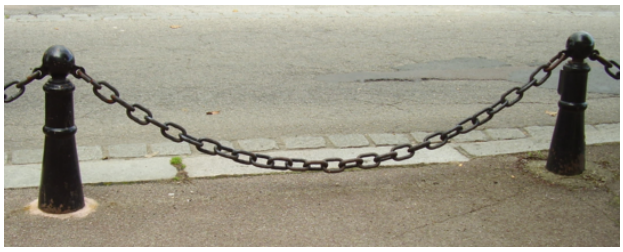
$$x^2 - y^2 = 1.$$

Similar parametric equations for the right branch:

$$x = \cosh\theta, \quad y = \sinh\theta.$$

What Is the Point of Hyperbolic Functions? Solution 5





Hyperbolic functions often arise from differential equations, for example in mechanics. Later we will use them in calculus.

Consider the forces on each point of a hanging string: horizontal tension T_0, tangent tension T, and weight $(0, -\lambda g s)$. Solving the differential equation gives a catenary.

In normalised coordinates:

$$y = \cosh x.$$

Equations for Hyperbolic Functions Solution 6

Hyperbolic sine ("shine"): $x \in \mathbb{R}$.

$\sinh x$

$$= \frac{e^x - e^{-x}}{2}.$$

Hyperbolic cosine ("cosh"): $x \in \mathbb{R}$.

$\cosh x$

$$= \frac{e^x + e^{-x}}{2}.$$

Hyperbolic tangent ("tanch"): $x \in \mathbb{R}$.

$\tanh x$

$$= \frac{\sinh x}{\cosh x}.$$

$$= \frac{e^{2x} - 1}{e^{2x} + 1}.$$

Hyperbolic secant ("setch"): $x \in \mathbb{R}$.

$\operatorname{sech} x$

$$= \frac{1}{\cosh x}.$$

$$= \frac{2}{e^x + e^{-x}}.$$

Hyperbolic cosecant ("cosetch"): $x \neq 0$.

$\operatorname{cosech} x$

$$= \frac{1}{\sinh x}.$$

$$= \frac{2}{e^x - e^{-x}}.$$

Hyperbolic cotangent ("coth"): $x \neq 0$.

$\coth x$

$$= \frac{1}{\tanh x}.$$

$$= \frac{e^{2x} + 1}{e^{2x} - 1}.$$

Equations for hyperbolic functions

Solution 7

Calculate $\sinh 3$ using both your calculator's hyperbolic function and the exponential formula.

Calculator tip: use the hyp function.

$$\sinh 3 = \frac{e^3 - e^{-3}}{2} \approx 10.02.$$

Write $\operatorname{cosech} 3$ in terms of e.

$$\operatorname{cosech} 3 = \frac{1}{\sinh 3} = \frac{2}{e^3 - e^{-3}}.$$

Solve $\sinh x = 5$.

$$\frac{e^x - e^{-x}}{2} = 5 \Rightarrow e^x - e^{-x} = 10$$

$$e^{2x} - 10e^x - 1 = 0$$

$$e^x = \frac{10 \pm \sqrt{104}}{2} \approx 10.099 \text{ or } -0.099.$$

Since $e^x > 0$, reject the negative root:

$$x = \ln(5 + \sqrt{26}) \approx 2.31.$$

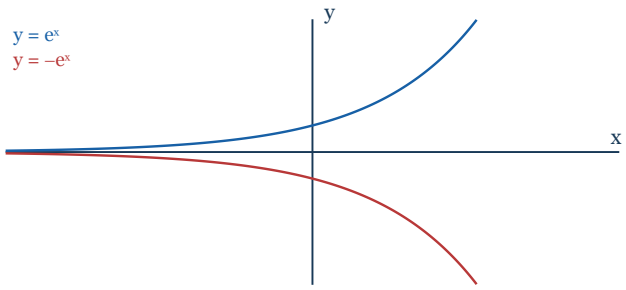
Find the exact value of $\tanh(\ln 4)$.

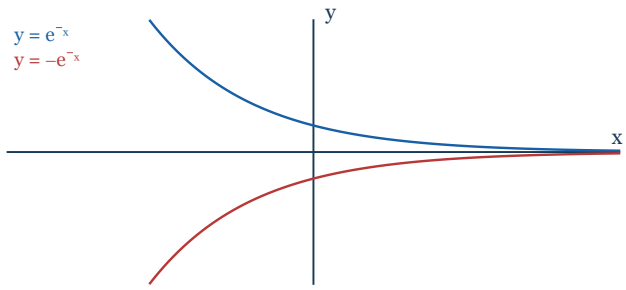
$$= \frac{e^{2\ln 4} - 1}{e^{2\ln 4} + 1} = \frac{e^{\ln 4^2} - 1}{e^{\ln 4^2} + 1}$$

$$= \frac{16 - 1}{16 + 1} = \frac{15}{17}.$$

Sketching Hyperbolic Functions

Solution 8





Take the average of e^x and $-e^{-x}$.

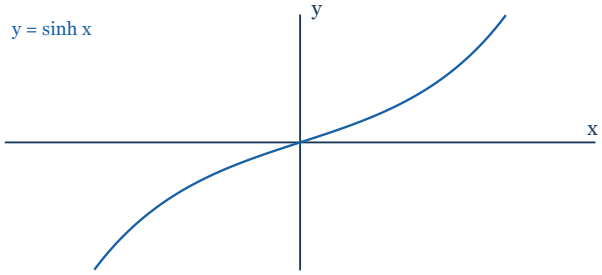
As $x \to \infty$, $e^{-x} \to 0$, so $\sinh x \sim \frac{1}{2}e^x$.

$$\sinh x = \frac{e^x - e^{-x}}{2}.$$

This is an odd function: $f(-x) = -f(x)$. Can you name others?

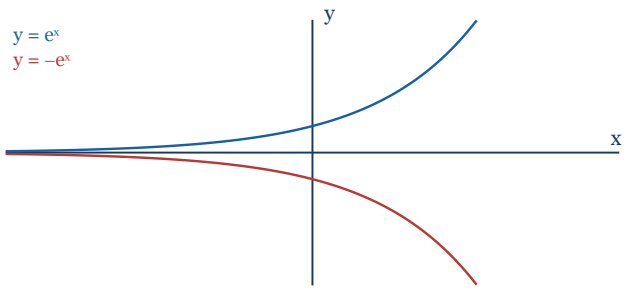
Other functions

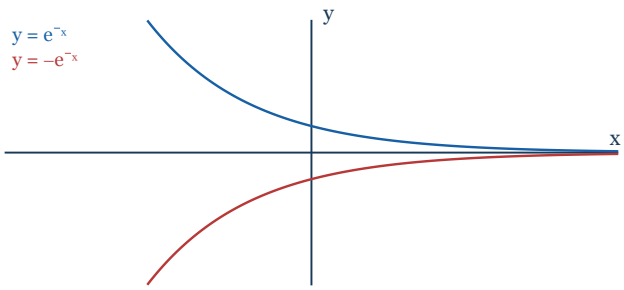
$\sin x$, x^3.



Sketching Hyperbolic Functions

Solution 9





Take the average of e^x and e^{-x}.

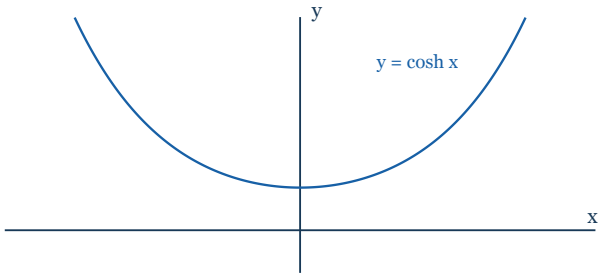
As $x \to \infty$, $e^{-x} \to 0$, so $\cosh x \sim \frac{1}{2}e^x$.

$$\cosh x = \frac{e^x + e^{-x}}{2}.$$

This is an even function: $f(-x) = f(x)$. Can you name others?

Other functions

$\cos x$, x^2.



Sketching Hyperbolic Functions

Solution 10

To sketch $y = \tanh x = \sinh x / \cosh x$, consider the usual features.

When $x = 0$:

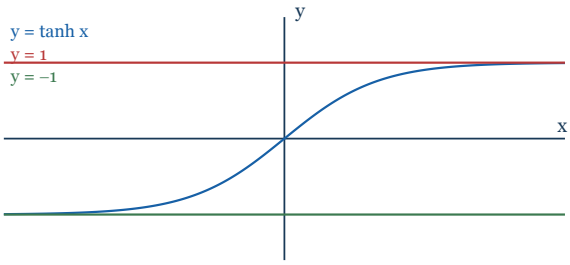
$$y = \frac{\sinh 0}{\cosh 0} = \frac{0}{1} = 0.$$

As $x \to \infty$:

$$\tanh x \sim \frac{\frac{1}{2}e^x}{\frac{1}{2}e^x} = 1.$$

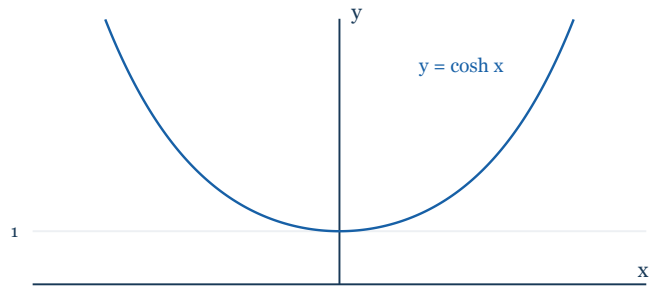
As $x \to -\infty$:

$$\tanh x \sim \frac{-\frac{1}{2}e^{-x}}{\frac{1}{2}e^{-x}} = -1.$$

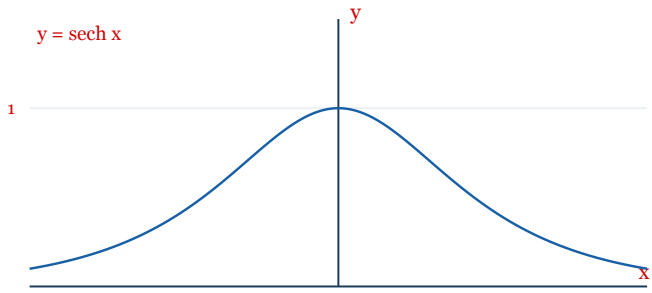


Test Your Understanding — Solution 11

Sketch $y = \operatorname{sech} x$.



Reciprocal sketch



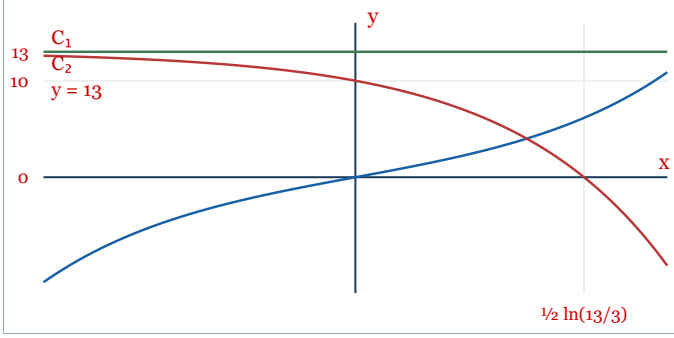
Take the reciprocal of each y-value.

FP3 June 2011 Q5(a)

$C_1 : y = 3\sinh 2x$, $C_2 : y = 13 - 3e^{2x}$.

Sketch both on one set of axes, giving any asymptote and the points where the curves cross the axes. (4 marks)

Exam sketch



Features

C_1 passes through $(0, 0)$; C_2 crosses at $(0, 10)$ and $(\frac{1}{2}\ln(13/3), 0)$. Its horizontal asymptote is $y = 13$.

Exercise 6A — Core Pure Book 2 Pages 122–123 — Solution 12

Exercise 6A — Core Pure Book 2, pages 122–123.

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Inverse Hyperbolic Functions

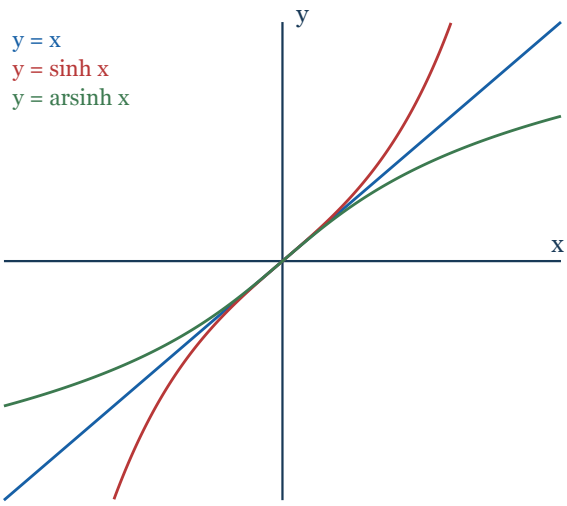
Solution 17

Each hyperbolic function has an inverse. Use the prefix "ar", as in arsinh (not arcsinh).

The six inverses are

$\operatorname{arsinh} x,\ \operatorname{arcosh} x,\ \operatorname{artanh} x,$

$\operatorname{arsech} x,\ \operatorname{arcosech} x,\ \operatorname{arcoth} x.$



$y = \sinh x.$

Reflect in $y = x$ to get $y = \operatorname{arsinh} x.$

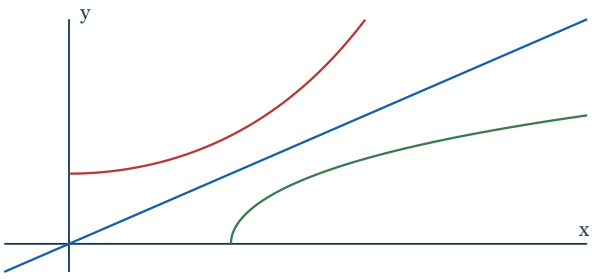
Inverse Hyperbolic Functions

Solution 18

Why is there a problem finding the inverse of $f(x) = \cosh x$?

A function has an inverse only if it is one-to-one. Unrestricted $\cosh x$ is many-to-one, so reflecting its whole graph would give a one-to-many relation.

Restrict $x \geq 0$. Then reflect in $y = x$ to get $y = \operatorname{arcosh} x$, with domain $x \geq 1$.



y = x
y = cosh x (x ≥ 0)
y = arcosh x

Inverse Hyperbolic Functions

Solution 19

Hyperbolic functions are exponential, so their inverses can be expressed using logarithms.

Prove $\operatorname{arsinh} x = \ln(x + \sqrt{x^2+1})$.

Quadratic

$$y = \operatorname{arsinh} x \Rightarrow x = \sinh y = \frac{e^y - e^{-y}}{2}$$

$$e^y - e^{-y} = 2x \Rightarrow e^{2y} - 2xe^y - 1 = 0.$$

As a quadratic in e^y: $a = 1, b = -2x, c = -1$.

Root

$$e^y = \frac{2x \pm \sqrt{4x^2+4}}{2} = x \pm \sqrt{x^2+1}.$$

Since $\sqrt{x^2+1} > |x|$ and $e^y > 0$, only the positive sign is possible.

Logarithm

$$y = \ln(x + \sqrt{x^2+1})$$

$$\operatorname{arsinh} x = \ln(x + \sqrt{x^2+1}).$$

Test Your Understanding

Solution 20

Prove $\operatorname{arcosh} x = \ln(x + \sqrt{x^2-1})$, $x \geq 1$.

Proof

$$y = \operatorname{arcosh} x \Rightarrow x = \cosh y = \frac{e^y + e^{-y}}{2}$$

$$e^y + e^{-y} = 2x \Rightarrow e^{2y} - 2xe^y + 1 = 0.$$

$$a = 1,\ b = -2x,\ c = 1$$

$$e^y = \frac{2x \pm \sqrt{4x^2-4}}{2} = x \pm \sqrt{x^2-1}.$$

Both roots are positive, but the principal inverse has $y \geq 0$. Thus

$$y = \ln(x + \sqrt{x^2-1}).$$

Show why the other logarithm is non-positive.

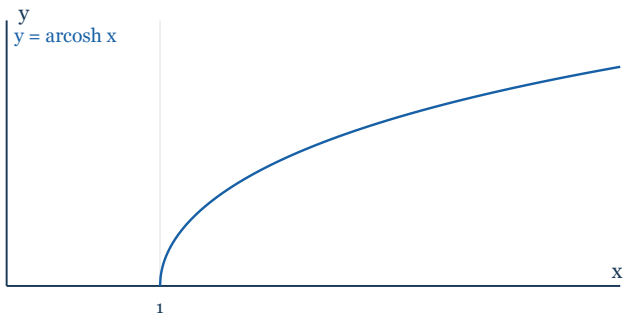
Other root

$$(x - \sqrt{x^2-1})(x + \sqrt{x^2-1}) = 1$$

$$x - \sqrt{x^2-1} = \frac{1}{x + \sqrt{x^2-1}}$$

$$\ln(x - \sqrt{x^2-1}) = -\ln(x + \sqrt{x^2-1}) \leq 0.$$

Equality occurs only at $x = 1$, where both branches give zero.



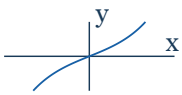
Summary So Far

Solution 21

$\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1})$ $\quad \operatorname{arcosh} x = \ln(x + \sqrt{x^2 - 1}),\ x \geq 1$ $\quad \operatorname{artanh} x = \frac{1}{2} \ln \frac{1+x}{1-x},\ |x| < 1$

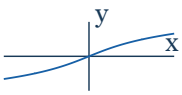
$y = \sinh x$

$x \in \mathbb{R}$



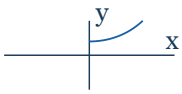
$y = \operatorname{arsinh} x$

$x \in \mathbb{R}$



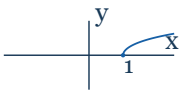
$y = \cosh x$

$x \geq 0$



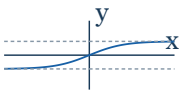
$y = \operatorname{arcosh} x$

$x \geq 1$



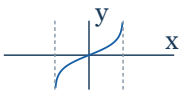
$y = \tanh x$

$x \in \mathbb{R}$



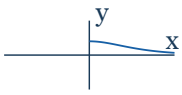
$y = \operatorname{artanh} x$

$|x| < 1$



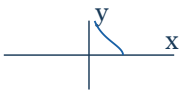
$y = \operatorname{sech} x$

$x \geq 0$



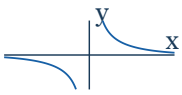
$y = \operatorname{arsech} x$

$0 < x \leq 1$



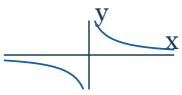
$y = \operatorname{cosech} x$

$x \neq 0$



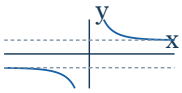
$y = \operatorname{arcosech} x$

$x \neq 0$



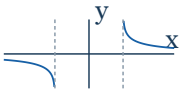
$y = \coth x$

$x \neq 0$



$y = \operatorname{arcoth} x$

$|x| > 1$



Exercise 6B — Core Pure Book 2 Pages 125

Solution 22

Exercise 6B — Core Pure Book 2, pages 125.

Exercise 6B — Core Pure Book 2, pages 125.

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Exercise 6B — Core Pure Book 2, pages 125.

Hyperbolic Identities

Solution 27

From Pure Year 1: $\sin^2 x + \cos^2 x = 1$. Are there similar hyperbolic identities?

Use the exponential definitions of sinh and cosh to prove the identities.

$\cosh^2 x - \sinh^2 x =$

1.

$\operatorname{sech}^2 x$

$= 1 - \tanh^2 x.$

$\operatorname{cosech}^2 x$

$= \coth^2 x - 1 \quad (x \neq 0).$

Hyperbolic Identities

Solution 28

We can similarly prove:

$\sinh(A \pm B) = \sinh A \cosh B \pm \cosh A \sinh B.$

This resembles the sine addition formula.

$\cosh(A \pm B) = \cosh A \cosh B \pm \sinh A \sinh B.$

Here the sign is $\pm$, unlike the $\mp$ in $\cos(A \pm B)$.

Prove that

$$\tanh(A \pm B) = \frac{\sinh(A \pm B)}{\cosh(A \pm B)}$$
$$= \frac{\tanh A \pm \tanh B}{1 \pm \tanh A \tanh B}.$$

For addition the denominator has a plus sign, unlike the ordinary tangent addition formula.

Osborn's Rule

Solution 29

Osborn's rule converts trigonometric identities:

1. Replace sin by sinh, and cos by cosh.
2. Negate any explicit or implied product of two sines.

$\sin A \sin B \longrightarrow$

$-\sinh A \sinh B.$

$\tan^2 A \longrightarrow$

$-\tanh^2 A$

Since $\tan^2 A = \sin^2 A / \cos^2 A$.

$\cos 2A = 2\cos^2 A - 1 \longrightarrow$

$\cosh 2A = 2\cosh^2 A - 1.$

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$\longrightarrow$

$$\tanh(A-B) = \frac{\tanh A - \tanh B}{1 - \tanh A \tanh B}.$$

Solving Equations

Solution 30

Use identities or exponential definitions. Solve for real x:

$6\sinh x - 2\cosh x = 7.$

$$6\left(\frac{e^x - e^{-x}}{2}\right) - 2\left(\frac{e^x + e^{-x}}{2}\right) = 7$$

$2e^x - 4e^{-x} = 7 \Rightarrow 2e^{2x} - 7e^x - 4 = 0$

$(2e^x + 1)(e^x - 4) = 0.$

Reject $e^x = -\frac{1}{2}$, so $x = \ln 4$.

Solve for real x:

$2\cosh^2 x - 5\sinh x = 5.$

Using $\cosh^2 x - \sinh^2 x = 1$:

$2(1 + \sinh^2 x) - 5\sinh x = 5$

$(2\sinh x + 1)(\sinh x - 3) = 0$

$\sinh x = -\frac{1}{2}$ or $\sinh x = 3.$

$x = \operatorname{arsinh}(-\frac{1}{2})$ or $x = \operatorname{arsinh} 3$

$$x = \ln\left(\frac{\sqrt{5}-1}{2}\right) \quad \text{or} \quad x = \ln(3+\sqrt{10}).$$

Further Examples

Solution 31

Pure Year 1 recap: if $\cos x = \frac{3}{5}$ and x is acute, find $\sin x$.

Recap

$\sin^2 x + \cos^2 x = 1$

$\sin x = \sqrt{1 - \frac{9}{25}} = \frac{4}{5}$.

Without the acute-angle assumption, either sign is possible.

If $\sinh x = \frac{3}{4}$, find exactly:

(a) $\cosh x$; (b) $\tanh x$; (c) $\sinh 2x$.

a

$\cosh^2 x - \sinh^2 x = 1$

$\cosh x = \sqrt{(\frac{3}{4})^2 + 1} = \frac{5}{4}$.

b

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{3}{5}.$$

c

$\sinh 2x = 2\sinh x \cosh x = \frac{15}{8}$.

Test Your Understanding

Solution 32

FP3 June 2009 Q1

Solve $7\operatorname{sech} x - \tanh x = 5$, giving answers as $\ln a$, where a is rational. (5 marks)

1

$$\frac{7}{\cosh x} - \frac{\sinh x}{\cosh x} = 5$$

$14 - (e^x - e^{-x}) = 5(e^x + e^{-x})$

$6e^x - 14 + 4e^{-x} = 0$

$3e^{2x} - 7e^x + 2 = 0$

$(3e^x - 1)(e^x - 2) = 0$

$x = \ln(\frac{1}{3})$ or $\ln 2$.

FP3 June 2014 (R) Q3

(a) Use exponential definitions to prove $\cosh^2 x - \sinh^2 x = 1$. (2 marks)

(b) Solve $2\sinh x + 7\cosh x = 9$ algebraically, giving natural logarithms. (5 marks)

2a

$\frac{1}{4}(e^x + e^{-x})^2 - \frac{1}{4}(e^x - e^{-x})^2$

$= \frac{1}{4}(e^{2x} + 2 + e^{-2x} - e^{2x} + 2 - e^{-2x}) = 1.$

2b

$(e^x - e^{-x}) + \frac{7}{2}(e^x + e^{-x}) = 9$

$9e^{2x} - 18e^x + 5 = 0$

$(3e^x - 1)(3e^x - 5) = 0$

$e^x = \frac{1}{3}$ or $\frac{5}{3}$

$x = \ln(\frac{1}{3})$ or $\ln(\frac{5}{3})$.

Exercise 6C — Core Pure Book 2 Pages 128–129 Solution 33

FP3 June 2011 Q5(b)

Solve $3\sinh 2x = 13 - 3e^{2x}$, giving the answer as $\frac{1}{2}\ln k$, where k is an integer. (5 marks)

Solution

$\frac{3}{2}(e^{2x} - e^{-2x}) = 13 - 3e^{2x}$

$9e^{4x} - 26e^{2x} - 3 = 0$

$(9e^{2x} + 1)(e^{2x} - 3) = 0$

$e^{2x} = -\frac{1}{9}$ or 3.

Reject the negative value:

$x = \frac{1}{2}\ln 3$.

Exercise 6C — Core Pure Book 2 Pages 128–129 Solution 34

Exercise 6C — Core Pure Book 2, pages 128–129.

Exercise 6C — Core Pure Book 2, pages 128–129.

Exercise 6C — Core Pure Book 2, pages 128–129.

Exercise 6C — Core Pure Book 2 Pages 128–129 Solution 37

Exercise 6C — Core Pure Book 2, pages 128–129.

Differentiating Hyperbolic Functions Solution 38

$\frac{d}{dx}\sinh x = \cosh x$

$\frac{d}{dx}\cosh x = \sinh x$

$\frac{d}{dx}\tanh x = \operatorname{sech}^2 x$

$\frac{d}{dx}\coth x = -\operatorname{cosech}^2 x.$

Memory tip: compare ordinary trig derivatives. The derivative of cosh has no minus; $(\operatorname{sech} x)' = -\operatorname{sech} x \tanh x$.

Prove $\frac{d}{dx}\sinh x = \cosh x$.

Proof

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\frac{d}{dx}\sinh x = \frac{e^x + e^{-x}}{2} = \cosh x.$$

Test Your Understanding

Solution 39

June 2014 (R) Q3

The curve has equation $y = \frac{1}{2}\ln(\coth x)$, $x > 0$.

Show that $dy/dx = -\operatorname{cosech} 2x$. (3 marks)

Hint: chain rule.

Chain rule

$$\frac{dy}{dx} = \frac{1}{2}\frac{1}{\coth x}(-\operatorname{cosech}^2 x)$$
$$= -\frac{1}{2\sinh x \cosh x} = -\frac{1}{\sinh 2x}$$
$$= -\operatorname{cosech} 2x.$$

Alternative

Alternatively:

$$e^{2y} = \coth x$$
$$2e^{2y}\frac{dy}{dx} = -\operatorname{cosech}^2 x$$
$$\frac{dy}{dx} = -\frac{\operatorname{cosech}^2 x}{2\coth x}$$
$$= -\frac{1}{2\sinh x \cosh x} = -\operatorname{cosech} 2x.$$

Inverse Hyperbolic Functions

Solution 40

$$\frac{d}{dx}\operatorname{arsinh} x = \frac{1}{\sqrt{x^2+1}}$$
$$\frac{d}{dx}\operatorname{arcosh} x = \frac{1}{\sqrt{x^2-1}} \quad (x > 1)$$
$$\frac{d}{dx}\operatorname{artanh} x = \frac{1}{1-x^2} \quad (|x| < 1).$$

Proof for arsinh:

$y = \operatorname{arsinh} x$

$\sinh y = x.$

$$\frac{dx}{dy} = \cosh y.$$

$$\frac{dx}{dy} = \sqrt{\sinh^2 y + 1} = \sqrt{x^2+1}.$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{x^2+1}}.$$

Find $\frac{d}{dx}\operatorname{artanh}(3x)$. Use $|x| < \frac{1}{3}$.

Example 1

By the chain rule:

$$\frac{d}{dx}\operatorname{artanh}(3x) = \frac{3}{1-(3x)^2} = \frac{3}{1-9x^2}.$$

Given $y = (\operatorname{arcosh} x)^2$, prove $(x^2-1)(dy/dx)^2 = 4y$, for $x > 1$.

Example 2

$$\frac{dy}{dx} = \frac{2\operatorname{arcosh} x}{\sqrt{x^2-1}}$$
$$\sqrt{x^2-1}\frac{dy}{dx} = 2\operatorname{arcosh} x$$
$$(x^2-1)\left(\frac{dy}{dx}\right)^2 = 4(\operatorname{arcosh} x)^2 = 4y.$$

Test Your Understanding

Solution 41

June 2009 Q4

Given $y = \operatorname{arsinh}(\sqrt{x})$, $x > 0$, find dy/dx as a simplified fraction. (3 marks)

$$\frac{dy}{dx} = \frac{\frac{1}{2}x^{-1/2}}{\sqrt{1+(\sqrt{x})^2}}$$
$$= \frac{1}{2\sqrt{x}\sqrt{1+x}} = \frac{1}{2\sqrt{x(1+x)}}.$$

June 2010 Q5

Given $y = (\operatorname{arcosh}(3x))^2$, $3x > 1$, show:

(a) $(9x^2-1)(dy/dx)^2 = 36y$. (5 marks)

(b) $(9x^2-1)d^2y/dx^2 + 9x\,dy/dx = 18$. (4 marks)

2a

$$\frac{dy}{dx} = \frac{6\operatorname{arcosh}(3x)}{\sqrt{9x^2-1}}$$
$$\sqrt{9x^2-1}\frac{dy}{dx} = 6\operatorname{arcosh}(3x)$$
$$(9x^2-1)\left(\frac{dy}{dx}\right)^2 = 36(\operatorname{arcosh}(3x))^2 = 36y.$$

2b

Differentiate (a):

$$18x\left(\frac{dy}{dx}\right)^2 + 2(9x^2-1)\frac{dy}{dx}\frac{d^2y}{dx^2} = 36\frac{dy}{dx}.$$

Since $x > \frac{1}{3}$, $dy/dx > 0$. Divide by $2dy/dx$:

$$(9x^2-1)\frac{d^2y}{dx^2} + 9x\frac{dy}{dx} = 18.$$

Using Maclaurin Expansions for Approximations

Solution 42

Textbook

(a) Show $\frac{d}{dx}\operatorname{arsinh} x = (1+x^2)^{-1/2}$. We did this earlier.

(b) Find the first two non-zero terms of the Maclaurin expansion.

Given

$$\operatorname{arsinh} x = \sum_{n=0}^{\infty} \frac{(-1)^n(2n)!}{2^{2n}(n!)^2}\frac{x^{2n+1}}{2n+1}, \quad |x| < 1,$$

(c) Find the coefficient of x^5.

(d) Use terms up to x^5 to approximate $\operatorname{arsinh} 0.5$.

(e) Calculate the percentage error.

b

$f(x) = \operatorname{arsinh} x, \quad f(0) = 0$

$f'(x) = (1+x^2)^{-1/2}, \quad f'(0) = 1$

$f''(x) = -x(1+x^2)^{-3/2}, \quad f''(0) = 0$

$f'''(x) = \dfrac{2x^2-1}{(1+x^2)^{5/2}}, \quad f'''(0) = -1.$

Continue until two non-zero terms appear:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

$$\operatorname{arsinh} x \approx x - \frac{x^3}{6}.$$

c

For $n = 2$:

$$\frac{(-1)^2 4!}{2^4(2!)^2}\frac{x^5}{5} = \frac{3}{40}x^5.$$

Coefficient: $3/40$.

d

$0.5 - \frac{1}{6}(0.5)^3 + \frac{3}{40}(0.5)^5$

$= 0.4815104167\ldots \approx 0.48151.$

e

$$\frac{0.4815104167\ldots - \operatorname{arsinh} 0.5}{\operatorname{arsinh} 0.5} \times 100$$
$\approx 0.062\%$ (3 d.p.).

Exercise 6D — Core Pure Book 2, pages 133–134.

Exercise 6D — Core Pure Book 2, pages 133–134.

Exercise 6D — Core Pure Book 2, pages 133–134.

Exercise 6D — Core Pure Book 2, pages 133–134.

Exercise 6D — Core Pure Book 2 Pages 133–134 Solution 47

Exercise 6D — Core Pure Book 2, pages 133–134.

Standard Integrals Solution 48

Compare the signs with ordinary trigonometric integrals. The cosech/coth results are additional context; the sech·tanh and cosech·coth results are not in the formula booklet. The arcsin and arctan results were covered in Chapter 3.

$$\int \sinh x \, dx = \cosh x + C. \qquad \int \operatorname{cosech} x \coth x \, dx = -\operatorname{cosech} x + C.$$

$$\int \cosh x \, dx = \sinh x + C. \qquad \int \frac{1}{\sqrt{1-x^2}} \, dx = \arcsin x + C, \quad |x| < 1.$$

$$\int \operatorname{sech}^2 x \, dx = \tanh x + C. \qquad \int \frac{1}{1+x^2} \, dx = \arctan x + C.$$

$$\int \operatorname{cosech}^2 x \, dx = -\coth x + C. \qquad \int \frac{1}{\sqrt{1+x^2}} \, dx = \operatorname{arsinh} x + C.$$

$$\int \operatorname{sech} x \tanh x \, dx = -\operatorname{sech} x + C. \qquad \int \frac{1}{\sqrt{x^2-1}} \, dx = \operatorname{arcosh} x + C, \quad x > 1.$$

Quickfire Examples — Do From Memory!

Solution 49

$\int \cosh(4x-1)\,dx$	$= \frac{1}{4}\sinh(4x-1)+C.$
$\int \sinh(\frac{2}{3}x)\,dx$	$= \frac{3}{2}\cosh(\frac{2}{3}x)+C.$
$\int \frac{3}{\sqrt{1+x^2}}\,dx$	$= 3\operatorname{arsinh} x + C.$
$\int \frac{4}{\sqrt{x^2-1}}\,dx$	$= 4\operatorname{arcosh} x + C.$
$\int \sinh 3x\,dx$	$= \frac{1}{3}\cosh 3x + C.$
$\int \frac{10}{\sqrt{x^2-1}}\,dx$	$= 10\operatorname{arcosh} x + C.$
$\int \frac{2}{\sqrt{1+x^2}}\,dx$	$= 2\operatorname{arsinh} x + C.$

Recall the reverse chain rule, $a \neq 0$:

$$\int f'(ax+b)\,dx = \frac{1}{a}f(ax+b)+C.$$

For example,

$$\int e^{3x+2}\,dx = \tfrac{1}{3}e^{3x+2}+C.$$

For the arcosh examples use $x > 1$.

$$\int \sinh x\,dx = \cosh x + C$$
$$\int \cosh x\,dx = \sinh x + C$$
$$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C \quad (|x|<1)$$
$$\int \frac{dx}{1+x^2} = \arctan x + C$$
$$\int \frac{dx}{\sqrt{1+x^2}} = \operatorname{arsinh} x + C$$
$$\int \frac{dx}{\sqrt{x^2-1}} = \operatorname{arcosh} x + C \quad (x>1).$$

Further Example

Solution 50

Textbook: $\int \frac{2+5x}{\sqrt{x^2+1}}\,dx$.

Working

$$= \int \frac{2}{\sqrt{x^2+1}}\,dx + \int \frac{5x}{\sqrt{x^2+1}}\,dx$$
$$= 2\int \frac{dx}{\sqrt{x^2+1}} + \int 5x(x^2+1)^{-1/2}\,dx$$
$$= 2\operatorname{arsinh} x + 5\sqrt{1+x^2} + C.$$

Integration strategy: if the numerator has multiple terms, split the fraction.

$$\int \sinh x\,dx = \cosh x + C$$
$$\int \cosh x\,dx = \sinh x + C$$
$$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C \quad (|x|<1)$$
$$\int \frac{dx}{1+x^2} = \arctan x + C$$
$$\int \frac{dx}{\sqrt{1+x^2}} = \operatorname{arsinh} x + C$$
$$\int \frac{dx}{\sqrt{x^2-1}} = \operatorname{arcosh} x + C \quad (x>1).$$

Integrating When Not Quite So Standard

Solution 51

Find $\int \cosh^5(2x)\sinh(2x)\,dx$.

Recall the "consider and scale" method used for $\int \sin^n x\cos x\,dx$ and $\int \cos^n x\sin x\,dx$.

Try $y = \cosh^6(2x)$:

$$\frac{dy}{dx} = 6\cosh^5(2x)\times 2\sinh(2x)$$

$$= 12\cosh^5(2x)\sinh(2x).$$

$$\therefore \int \cosh^5(2x)\sinh(2x)\,dx = \tfrac{1}{12}\cosh^6(2x) + C.$$

Find $\int \tanh x\,dx$.

As for ordinary tan, first express the function as a quotient.

$$\int \tanh x\,dx = \int \frac{\sinh x}{\cosh x}\,dx$$

$$= \ln|\cosh x| + C.$$

There is no minus sign: cosh differentiates to positive sinh. Also $\cosh x > 0$, so the modulus can be omitted.

Using Identities

Solution 52

Find $\int \cosh^2(3x)\,dx$.

Use the hyperbolic double-angle formula.

$$= \int \left(\tfrac{1}{2} + \tfrac{1}{2}\cosh 6x\right)\,dx$$

$$= \tfrac{1}{2}x + \tfrac{1}{12}\sinh 6x + C.$$

Find $\int \sinh^3 x\,dx$.

This method works for small odd powers of sinh or cosh.

$$= \int \sinh^2 x\sinh x\,dx$$

$$= \int (\cosh^2 x - 1)\sinh x\,dx$$

$$= \int (\cosh^2 x\sinh x - \sinh x)\,dx$$

$$= \tfrac{1}{3}\cosh^3 x - \cosh x + C.$$

When That Does Not Work...

Solution 53

When a familiar trigonometric technique is unhelpful, try replacing hyperbolic functions with their exponential definitions.

Find $\int e^{2x} \sinh x \, dx$.

$$= \int e^{2x} \left(\frac{e^x - e^{-x}}{2} \right) dx$$

$$= \tfrac{1}{2} \int (e^{3x} - e^x) \, dx$$

$$= \tfrac{1}{6}(e^{3x} - 3e^x) + C.$$

Integration by parts also works, but needs much more working.

Find $\int \operatorname{sech} x \, dx$.

$$= \int \frac{2}{e^x + e^{-x}} \, dx = \int \frac{2e^x}{e^{2x} + 1} \, dx.$$

Let $u = e^x$, so $du = e^x \, dx$:

$$= \int \frac{2}{u^2 + 1} \, du = 2 \arctan u + C$$

$$= 2 \arctan(e^x) + C.$$

Hyperbolic Substitutions

Solution 54

For $a > 0$, choose sensible substitutions.

Recall $\sin^2 \theta + \cos^2 \theta = 1$, $1 + \tan^2 \theta = \sec^2 \theta$, and $1 + \sinh^2 u = \cosh^2 u$.

$\int \frac{dx}{\sqrt{a^2+x^2}}$

Use $x = a \sinh u$. The denominator becomes $a \cosh u$ and $dx = a \cosh u \, du$, so they cancel.

With $x = a \tan \theta$, the denominator gives $a \sec \theta$ but $dx = a \sec^2 \theta \, d\theta$, leaving an extra sec factor.

$\int \frac{dx}{\sqrt{x^2-a^2}}$, $x > a$

Use $x = a \cosh u$, $u > 0$.

Show $\int \frac{dx}{\sqrt{x^2-a^2}} = \operatorname{arcosh}(x/a) + C$, $x > a$.

Proof

$$x = a \cosh u, \quad dx = a \sinh u \, du$$

$$\int \frac{a \sinh u}{\sqrt{a^2 \cosh^2 u - a^2}} \, du$$

$$= \frac{1}{a} \int \frac{a \sinh u}{\sqrt{\cosh^2 u - 1}} \, du = \int \frac{\sinh u}{\sinh u} \, du$$

$$= u + C = \operatorname{arcosh}(x/a) + C.$$

$$\int \frac{dx}{\sqrt{a^2 + x^2}} = \operatorname{arsinh}(x/a) + C$$

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \operatorname{arcosh}(x/a) + C, \; x > a.$$

Hyperbolic Substitutions

Solution 55

Textbook: show

$$\int_5^8 \frac{dx}{\sqrt{x^2-16}} = \ln\left(\frac{2+\sqrt{3}}{2}\right).$$

Working

$$= [\operatorname{arcosh}(x/4)]_5^8$$
$$= \operatorname{arcosh} 2 - \operatorname{arcosh}(5/4)$$
$$= \ln(2+\sqrt{3}) - \ln\left(\tfrac{5}{4} + \sqrt{\tfrac{9}{16}}\right)$$
$$= \ln(2+\sqrt{3}) - \ln 2$$
$$= \ln\left(\frac{2+\sqrt{3}}{2}\right).$$

For $a > 0$:

$$\int \frac{dx}{\sqrt{a^2+x^2}} = \operatorname{arsinh}(x/a) + C$$
$$\int \frac{dx}{\sqrt{x^2-a^2}} = \operatorname{arcosh}(x/a) + C,\ x > a$$
$$\operatorname{arsinh} x = \ln(x + \sqrt{x^2+1})$$
$$\operatorname{arcosh} x = \ln(x + \sqrt{x^2-1}),\ x \geq 1.$$

Harder Example

Solution 56

Show $\int \sqrt{1+x^2}\,dx = \frac{1}{2}\operatorname{arsinh} x + \frac{1}{2}x\sqrt{1+x^2} + C$.

Hint: use a sensible substitution. Recall $\sin^2\theta + \cos^2\theta = 1$, $1 + \tan^2\theta = \sec^2\theta$, $1 + \sinh^2 u = \cosh^2 u$.

Trigonometric substitution

Try $x = \tan\theta$, $-\pi/2 < \theta < \pi/2$:

$$dx = \sec^2\theta\,d\theta, \quad I = \int \sec^3\theta\,d\theta.$$

Integrate by parts with $u = \sec\theta$, $dv/d\theta = \sec^2\theta$, hence $du/d\theta = \sec\theta\tan\theta$, $v = \tan\theta$:

$$I = \sec\theta\tan\theta - \int \sec\theta\tan^2\theta\,d\theta$$
$$= \sec\theta\tan\theta - \int \sec\theta(\sec^2\theta - 1)\,d\theta$$
$$= \sec\theta\tan\theta - I + \int \sec\theta\,d\theta.$$
$$2I = \sec\theta\tan\theta + \ln|\sec\theta + \tan\theta| + C$$
$$I = \tfrac{1}{2}\sec\theta\tan\theta + \tfrac{1}{2}\ln|\sec\theta + \tan\theta| + C$$
$$= \tfrac{1}{2}x\sqrt{1+x^2} + \tfrac{1}{2}\ln|\sqrt{1+x^2} + x| + C$$
$$= \tfrac{1}{2}x\sqrt{1+x^2} + \tfrac{1}{2}\operatorname{arsinh} x + C.$$

Hyperbolic substitution

Try $x = \sinh u$:

$$dx = \cosh u\,du$$
$$I = \int \cosh u \cosh u\,du = \int \cosh^2 u\,du$$
$$= \tfrac{1}{2}\int (1 + \cosh 2u)\,du$$
$$= \tfrac{1}{2}\left(u + \tfrac{1}{2}\sinh 2u\right) + C$$
$$= \tfrac{1}{2}u + \tfrac{1}{2}\sinh u\cosh u + C$$
$$= \tfrac{1}{2}\operatorname{arsinh} x + \tfrac{1}{2}x\sqrt{1+x^2} + C.$$

This substitution gives the shorter method.

Test Your Understanding So Far

Solution 57

June 2013 Q2

(a) Find $\int \frac{dx}{\sqrt{4x^2+9}}$. (2 marks)

(b) Hence evaluate $\int_{-3}^{3} \frac{dx}{\sqrt{4x^2+9}}$, in the form $k\ln(a+b\sqrt{5})$, with integers a, b and constant k. (3 marks)

Hint: factor out $1/\sqrt{4}$, as in Chapter 3.

a

$$\frac{1}{2}\int \frac{dx}{\sqrt{x^2+(3/2)^2}}$$

$$= \tfrac{1}{2}\operatorname{arsinh}(2x/3) + C.$$

Equivalently, $\frac{1}{2}\ln\left|2x+\sqrt{4x^2+9}\right| + C.$

b

$$\tfrac{1}{2}\ln(6+\sqrt{45}) - \tfrac{1}{2}\ln(-6+\sqrt{45})$$

$$= \tfrac{1}{2}\ln\frac{6+\sqrt{45}}{-6+\sqrt{45}}$$

$$= \tfrac{1}{2}\ln\left(\frac{6+\sqrt{45}}{-6+\sqrt{45}}\frac{6+\sqrt{45}}{6+\sqrt{45}}\right)$$

$$= \tfrac{1}{2}\ln\frac{(6+\sqrt{45})^2}{9}$$

$$= \tfrac{1}{2}\ln(9+4\sqrt{5}) = \ln(2+\sqrt{5}).$$

Alternatively use symmetry:

$$2\left[\tfrac{1}{2}\ln(2x+\sqrt{4x^2+9})\right]_0^3$$

$$= \ln(6+\sqrt{45}) - \ln 3 = \ln(2+\sqrt{5}).$$

Use a hyperbolic substitution to evaluate

$$\int_0^6 \frac{x^3}{\sqrt{x^2+9}}\,dx.$$

Further example

Using $x = 3\sinh u$ gives $18(\sqrt{5}+1)$.

Integrating by Completing the Square

Solution 58

Determine $\int \frac{dx}{x^2-8x+8}$.

Complete the square, then use a standard result. A substitution makes the form clearer.

Working

$$= \int \frac{dx}{(x-4)^2-8}.$$

Let $u = x-4$, so $du = dx$:

$$= \int \frac{du}{u^2-8} = \frac{1}{2\sqrt{8}}\ln\left|\frac{u-\sqrt{8}}{u+\sqrt{8}}\right| + C$$

$$= \frac{\sqrt{2}}{8}\ln\left|\frac{x-4-2\sqrt{2}}{x-4+2\sqrt{2}}\right| + C.$$

Standard results (formula booklet):

For $a > 0$:

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin(x/a) + C, \quad |x| < a$$

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan(x/a) + C$$

$$\int \frac{dx}{\sqrt{a^2+x^2}} = \operatorname{arsinh}(x/a) + C$$

$$\int \frac{dx}{\sqrt{x^2-a^2}} = \operatorname{arcosh}(x/a) + C, \quad x > a$$

$$\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + C$$

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C.$$

Further Example

Solution 59

Determine $\int \frac{dx}{\sqrt{12x+2x^2}}$ for $x > 0$.

Working

$2x^2 + 12x = 2(x^2 + 6x) = 2((x+3)^2 - 9).$

$$\int \frac{dx}{\sqrt{2((x+3)^2-9)}} = \frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{(x+3)^2-9}}.$$

Let $u = x + 3$, so $du = dx$:

$$= \frac{1}{\sqrt{2}} \int \frac{du}{\sqrt{u^2-9}} = \frac{1}{\sqrt{2}} \operatorname{arcosh}(u/3) + C$$

$$= \frac{1}{\sqrt{2}} \operatorname{arcosh}\left(\frac{x+3}{3}\right) + C.$$

Standard results (formula booklet):

For $a > 0$:

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin(x/a) + C, \quad |x| < a$$

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a} \arctan(x/a) + C$$

$$\int \frac{dx}{\sqrt{a^2+x^2}} = \operatorname{arsinh}(x/a) + C$$

$$\int \frac{dx}{\sqrt{x^2-a^2}} = \operatorname{arcosh}(x/a) + C, \quad x > a$$

$$\int \frac{dx}{a^2-x^2} = \frac{1}{2a} \ln\left|\frac{a+x}{a-x}\right| + C$$

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a} \ln\left|\frac{x-a}{x+a}\right| + C.$$

Test Your Understanding

Solution 60

June 2014 (R) Q2

$9x^2 + 6x + 5 = a(x+b)^2 + c.$

(a) Find a, b, c. (3 marks)

Hence, or otherwise, find:

(b) $\int \frac{dx}{9x^2+6x+5}$. (2 marks)

(c) $\int \frac{dx}{\sqrt{9x^2+6x+5}}$. (2 marks)

a

$9x^2 + 6x + 5 = 9(x + \frac{1}{3})^2 + 4.$

$a = 9, \quad b = \frac{1}{3}, \quad c = 4.$

b

$$\int \frac{dx}{9(x+\frac{1}{3})^2+4} = \frac{1}{6} \arctan\left(\frac{3x+1}{2}\right) + C.$$

c

$$\int \frac{dx}{\sqrt{9(x+\frac{1}{3})^2+4}} = \frac{1}{3} \operatorname{arsinh}\left(\frac{3x+1}{2}\right) + C.$$

Exercise 6E — Core Pure Book 2, pages 140–142.

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